1.3 Polar Form of Complex Numbers

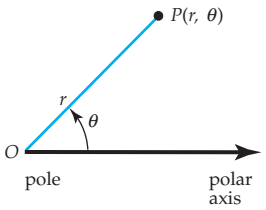


FIGURE 1.3.1 Polar coordinates

INTRODUCTION Recall from calculus that a point P in the plane whose rectangular coordinates are (x, y) can also be described in terms of **polar coordinates**. The polar coordinate system, invented by Isaac Newton, consists of point O called the **pole** and the horizontal half-line emanating from the pole called the **polar axis**. If r is a directed distance from the pole to P and θ is an angle of inclination (in radians) measured from the polar axis to the line OP, then the point can be described by the ordered pair (r, θ), called the polar coordinates of P. See FIGURE 1.3.1.

A brief review of polar coordinates can be found in Section 1.3 of the student study guide. ▶

Polar Form Suppose, as shown in FIGURE 1.3.2, that a polar coordinate system is superimposed on the complex plane with the polar axis coinciding with the positive x-axis and the pole O at the origin. Then x, y, r, and θ are related by $x = r\cos\theta$, $y = r\sin\theta$. These equations enable us to express a nonzero complex number $z = x + iy$ as $z = (r\cos\theta) + i(r\sin\theta)$ or

$$z = r(\cos\theta + i\sin\theta). \tag{1}$$

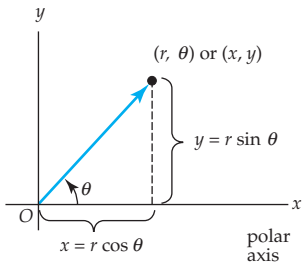


FIGURE 1.3.2 Polar coordinates in the complex plane

We say that (1) is the **polar form** or **polar representation** of the complex number z. Again, from Figure 1.3.2 we see that the coordinate r can be interpreted as the distance from the origin to the point (x, y). In other words, we shall adopt the convention that *r is never negative** so that we can take r to be the modulus of z, that is, $r = |z|$. The angle θ of inclination of the vector z, which *will always be measured in radians* from the positive real axis, is positive when measured counterclockwise and negative when measured clockwise. The angle θ is called an **argument** of z and is denoted by $\theta = \arg(z)$. An argument θ of a complex number must satisfy the equations $\cos\theta = x/r$ and $\sin\theta = y/r$. An argument of a complex number z is not unique since $\cos\theta$ and $\sin\theta$ are 2π-periodic; in other words, if θ_0 is an argument of z, then necessarily the angles $\theta_0 \pm 2\pi$, $\theta_0 \pm 4\pi$, ... are also arguments of z. In practice we use $\tan\theta = y/x$ to find θ. However, because $\tan\theta$ is π-periodic, some care must be exercised in using the last equation. A calculator will give only angles satisfying $-\pi/2 < \tan^{-1}(y/x) < \pi/2$, that is, angles in the first and fourth quadrants. We have to choose θ consistent with the quadrant in which z is located; this may require adding or subtracting π to $\tan^{-1}(y/x)$ when appropriate. The following example illustrates how this is done.

Be careful using $\tan^{-1}(y/x)$ ▶

EXAMPLE 1 A Complex Number in Polar Form

Express $-\sqrt{3} - i$ in polar form.

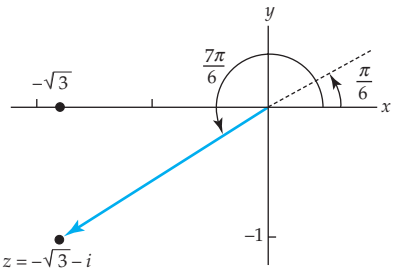


FIGURE 1.3.3 $\arg(-\sqrt{3} - i)$

Solution With $x = -\sqrt{3}$ and $y = -1$ we obtain $r = |z| = \sqrt{(-\sqrt{3})^2 + (-1)^2} = 2$. Now $y/x = -1/(-\sqrt{3}) = 1/\sqrt{3}$, and so a calculator gives $\tan^{-1}(1/\sqrt{3}) = \pi/6$, which is an angle whose terminal side is in the first quadrant. But since the point $-\sqrt{3} - i$ lies in the third quadrant, we take the solution of $\tan\theta = -1/(-\sqrt{3}) = 1/\sqrt{3}$ to be $\theta = \arg(z) = \pi/6 + \pi = 7\pi/6$. See FIGURE 1.3.3. It follows from (1) that a polar form of the number is

$$z = 2\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right). \tag{2}$$

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*In general, in the polar description (r, θ) of a point P in the Cartesian plane, we can have $r \geq 0$ or $r < 0$.

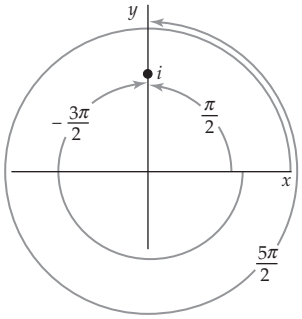


FIGURE 1.3.4 Some arguments of i

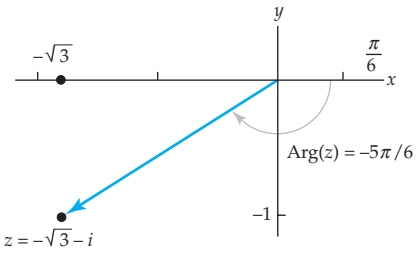


FIGURE 1.3.5 Principal argument $z = -\sqrt{3} - i$

Principal Argument The symbol $\arg(z)$ represents a set of values, but the choice of an argument θ of a complex number that lies in the interval $-\pi < \theta \leq \pi$ is called the **principal value** of $\arg(z)$ or the **principal argument** of z. The principal argument of z is *unique* and is represented by the symbol $\text{Arg}(z)$, that is,

$$-\pi < \text{Arg}(z) \leq \pi.$$

For example, if $z = i$, we see in **FIGURE 1.3.4** that some values of $\arg(i)$ are $\pi/2,\ 5\pi/2,\ -3\pi/2$, and so on, but $\text{Arg}(i) = \pi/2$. Similarly, we see from **FIGURE 1.3.5** that the argument of $-\sqrt{3} - i$ that lies in the interval $(-\pi,\ \pi]$, the principal argument of z, is $\text{Arg}(z) = \pi/6 - \pi = -5\pi/6$. Using $\text{Arg}(z)$ we can express the complex number in (2) in the alternative polar form:

$$z = 2\left[\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right].$$

In general, $\arg(z)$ and $\text{Arg}(z)$ are related by

$$\arg(z) = \text{Arg}(z) + 2n\pi, \quad n = 0,\ \pm 1,\ \pm 2, \ldots. \tag{3}$$

For example, $\arg(i) = \dfrac{\pi}{2} + 2n\pi$. For the choices $n = 0$ and $n = -1$, (3) gives $\arg(i) = \text{Arg}(i) = \pi/2$ and $\arg(i) = -3\pi/2$, respectively.

Multiplication and Division The polar form of a complex number is especially convenient when multiplying or dividing two complex numbers. Suppose

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1) \quad \text{and} \quad z_2 = r_2(\cos\theta_2 + i\sin\theta_2),$$

where θ_1 and θ_2 are any arguments of z_1 and z_2, respectively. Then

$$z_1 z_2 = r_1 r_2\left[\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 + i\left(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2\right)\right] \tag{4}$$

and, for $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\left[\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2 + i\left(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2\right)\right]. \tag{5}$$

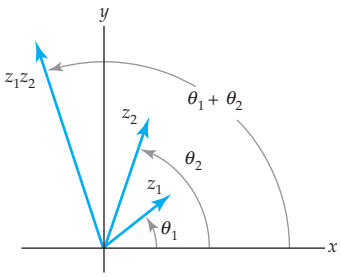


FIGURE 1.3.6 $\arg(z_1z_2) = \theta_1 + \theta_2$

From the addition formulas* for the cosine and sine, (4) and (5) can be rewritten as

$$z_1z_2 = r_1r_2\,[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)] \tag{6}$$

and

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\,[\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)]. \tag{7}$$

Inspection of the expressions in (6) and (7) and **FIGURE 1.3.6** shows that the lengths of the two vectors z_1z_2 and z_1/z_2 are the product of the lengths of z_1 and z_2 and the quotient of the lengths of z_1 and z_2, respectively. See (3) of Section 1.2. Moreover, the arguments of z_1z_2 and z_1/z_2 are given by

$$\arg(z_1z_2) = \arg(z_1) + \arg(z_2) \quad \text{and} \quad \arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2). \tag{8}$$

EXAMPLE 2 Argument of a Product and of a Quotient

We have just seen that for $z_1 = i$ and $z_2 = -\sqrt{3} - i$ that $\text{Arg}(z_1) = \pi/2$ and $\text{Arg}(z_2) = -5\pi/6$, respectively. Thus arguments for the product and quotient

$$z_1z_2 = i(-\sqrt{3} - i) = 1 - \sqrt{3}i \quad \text{and} \quad \frac{z_1}{z_2} = \frac{i}{-\sqrt{3} - i} = -\frac{1}{4} - \frac{\sqrt{3}}{4}i$$

can be obtained from (8):

$$\arg(z_1z_2) = \frac{\pi}{2} + \left(-\frac{5\pi}{6}\right) = -\frac{\pi}{3} \quad \text{and} \quad \arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2} - \left(-\frac{5\pi}{6}\right) = \frac{4\pi}{3}.$$

Additional values of $\arg(z_1z_2)$ and $\arg(z_1/z_2)$ can be found by adding integer multiples of 2π to $= \pi/3$ or $4\pi/3$, respectively.

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Integer Powers of z We can find integer powers of a complex number z from the results in (6) and (7). For example, if $z = r(\cos\theta + i\sin\theta)$, then with $z_1 = z_2 = z$, (6) gives

$$z^2 = r^2\,[\cos(\theta + \theta) + i\sin(\theta + \theta)] = r^2\,(\cos 2\theta + i\sin 2\theta)\,.$$

Since $z^3 = z^2z$, it then follows that

$$z^3 = r^3\,(\cos 3\theta + i\sin 3\theta)\,,$$

and so on. In addition, if we take $\arg(1) = 0$, then (7) gives

$$z^{-1} = \frac{1}{z} = \frac{1}{r}\,[\cos(0 - \theta) + i\,\sin(0 - \theta)] = r^{-1}\,[\cos(-\theta) + i\sin(-\theta)]\,.$$

Continuing in this manner, we obtain a formula for the nth power of z for any integer n:

$$z^n = r^n\,(\cos n\theta + i\sin n\theta). \tag{9}$$

When $n = 0$, we get the familiar result $z^0 = 1$.

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*$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$ and $\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$

EXAMPLE 3 Power of a Complex Number

Compute z^3 for $z = -\sqrt{3} - i$.

Solution In (2) of Example 1 we saw that a polar form of the given number is $z = 2[\cos(7\pi/6) + i\sin(7\pi/6)]$. Using (9) with $r = 2$, $\theta = 7\pi/6$, and $n = 3$ we get

$$\left(-\sqrt{3} - i\right)^3 = 2^3\left[\cos\left(3\frac{7\pi}{6}\right) + i\sin\left(3\frac{7\pi}{6}\right)\right] = 8\left[\cos\frac{7\pi}{2} + i\sin\frac{7\pi}{2}\right] = -8i$$

since $\cos(7\pi/2) = 0$ and $\sin(7\pi/2) = -1$. ☰

Note in Example 3, if we also want the value of z^{-3}, then we could proceed in two ways: either find the reciprocal of $z^3 = -8i$ or use (9) with $n = -3$.

de Moivre's Formula When $z = \cos\theta + i\sin\theta$, we have $|z| = r = 1$, and so (9) yields

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta. \tag{10}$$

This last result is known as **de Moivre's formula** and is useful in deriving certain trigonometric identities involving $\cos n\theta$ and $\sin n\theta$. See Problems 33 and 34 in Exercises 1.3.

EXAMPLE 4 de Moivre's Formula

From (10), with $\theta = \pi/6$, $\cos\theta = \sqrt{3}/2$, and $\sin\theta = 1/2$:

$$\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^3 = \cos 3\theta + i\sin 3\theta = \cos\left(3\cdot\frac{\pi}{6}\right) + i\sin\left(3\cdot\frac{\pi}{6}\right)$$
$$= \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i.$$

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Remarks COMPARISON WITH REAL ANALYSIS

(*i*) Observe in Example 2 that even though we used the principal arguments of z_1 and z_2 that $\arg(z_1/z_2) = 4\pi/3 \neq \text{Arg}(z_1/z_2)$. Although (8) is true for any arguments of z_1 and z_2, *it is not true*, in general, that $\text{Arg}(z_1z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$ and $\text{Arg}(z_1/z_2) = \text{Arg}(z_1) - \text{Arg}(z_2)$. See Problems 37 and 38 in Exercises 1.3.

(*ii*) An argument can be assigned to any *nonzero* complex number z. However, for $z = 0$, $\arg(z)$ cannot be defined in any way that is meaningful.

(*iii*) If we take $\arg(z)$ from the interval $(-\pi,\ \pi]$, the relationship between a complex number z and its argument is single-valued; that is, every nonzero complex number has precisely *one* angle in $(-\pi,\ \pi]$. But there is nothing special about the interval $(-\pi,\ \pi]$; we also establish a single-valued relationship by using the interval $(0,\ 2\pi]$ to define the principal value of the argument of z. For the interval $(-\pi,\ \pi]$, the negative real axis is analogous to a barrier that we agree not to cross; the technical name for this barrier is a **branch cut**. If we use $(0,\ 2\pi]$, the branch cut is the positive real axis. The concept of a branch cut is important and will be examined in greater detail when we study functions in Chapters 3 and 4.

(*iv*) The "cosine *i* sine" part of the polar form of a complex number is sometimes abbreviated cis. That is,

$$z = r(\cos\theta + i\sin\theta) = r\,\text{cis}\,\theta.$$

This notation, used mainly in engineering, will not be used in this text.

EXERCISES 1.3

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–10, write the given complex number in polar form first using an argument $\theta \neq \text{Arg}(z)$ and then using $\theta = \text{Arg}(z)$.

1. 2

2. -10

3. $-3i$

4. $6i$

5. $1 + i$

6. $5 - 5i$

7. $-\sqrt{3} + i$

8. $-2 - 2\sqrt{3}i$

9. $\dfrac{3}{-1+i}$

10. $\dfrac{12}{\sqrt{3}+i}$

In Problems 11 and 12, use a calculator to write the given complex number in polar form first using an argument $\theta \neq \text{Arg}(z)$ and then using $\theta = \text{Arg}(z)$.

11. $-\sqrt{2} + \sqrt{7}i$

12. $-12 - 5i$

In Problems 13 and 14, write the complex number whose polar coordinates $(r,\ \theta)$ are given in the form $a + ib$. Use a calculator if necessary.

13. $(4,\ -5\pi/3)$

14. $(2,\ 2)$

In Problems 15–18, write the complex number whose polar form is given in the form $a + ib$. Use a calculator if necessary.

15. $z = 5\left(\cos\dfrac{7\pi}{6} + i\sin\dfrac{7\pi}{6}\right)$

16. $z = 8\sqrt{2}\left(\cos\dfrac{11\pi}{4} + i\sin\dfrac{11\pi}{4}\right)$

17. $z = 6\left(\cos\dfrac{\pi}{8} + i\sin\dfrac{\pi}{8}\right)$

18. $z = 10\left(\cos\dfrac{\pi}{5} + i\sin\dfrac{\pi}{5}\right)$

In Problems 19 and 20, use (6) and (7) to find z_1z_2 and z_1/z_2. Write the number in the form $a + ib$.

19. $z_1 = 2\left(\cos\dfrac{\pi}{8} + i\sin\dfrac{\pi}{8}\right),\ z_2 = 4\left(\cos\dfrac{3\pi}{8} + i\sin\dfrac{3\pi}{8}\right)$

20. $z_1 = \sqrt{2}\left(\cos\dfrac{\pi}{4} + i\sin\dfrac{\pi}{4}\right),\ z_2 = \sqrt{3}\left(\cos\dfrac{\pi}{12} + i\sin\dfrac{\pi}{12}\right)$

In Problems 21–24, write each complex number in polar form. Then use either (6) or (7) to obtain the polar form of the given number. Finally, write the polar form in the form $a + ib$.

21. $(3 - 3i)(5 + 5\sqrt{3}i)$

22. $(4 + 4i)(-1 + i)$

23. $\dfrac{-i}{1+i}$

24. $\dfrac{\sqrt{2} + \sqrt{6}i}{-1 + \sqrt{3}i}$

In Problems 25–30, use (9) to compute the indicated powers. Write the complex number in polar form and then in the form $a + ib$.

25. $(1 + \sqrt{3}i)^9$

26. $(2 - 2i)^5$

27. $\left(\frac{1}{2} + \frac{1}{2}i\right)^{10}$

28. $(-\sqrt{2} + \sqrt{6}i)^4$

29. $\left[\left(\sqrt{2}\cos\dfrac{\pi}{8} + i\sqrt{2}\sin\dfrac{\pi}{8}\right)\right]^{12}$

30. $\left[\sqrt{3}\left(\cos\dfrac{2\pi}{9} + i\sin\dfrac{2\pi}{9}\right)\right]^{6}$

In Problems 31 and 32, write the given complex number in polar form and then in the form $a + ib$.

31. $\left(\cos\frac{\pi}{9} + i\sin\frac{\pi}{9}\right)^{12}\left[2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right]^{5}$

32. $\dfrac{\left[8\left(\cos\frac{3\pi}{8} + i\sin\frac{3\pi}{8}\right)\right]^{3}}{\left[2\left(\cos\frac{\pi}{16} + i\sin\frac{\pi}{16}\right)\right]^{10}}$

33. Use de Moivre's formula (10) with $n = 2$ to find trigonometric identities for $\cos 2\theta$ and $\sin 2\theta$.

34. Use de Moivre's formula (10) with $n = 3$ to find trigonometric identities for $\cos 3\theta$ and $\sin 3\theta$.

In Problems 35 and 36, find a positive integer n for which the equality holds.

35. $\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{n} = -1$ **36.** $\left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right)^{n} = 1$

37. For the complex numbers $z_1 = -1$ and $z_2 = 5i$, verify that:

(a) $\text{Arg}(z_1 z_2) \neq \text{Arg}(z_1) + \text{Arg}(z_2)$

(b) $\text{Arg}(-z_2/z_1) \neq \text{Arg}(-z_2) - \text{Arg}(z_1)$.

38. For the complex numbers given in Problem 37, verify that:

(a) $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$

(b) $\arg(-z_2/z_1) = \arg(-z_2) - \arg(z_1)$.

Focus on Concepts

39. Suppose that $z = r(\cos\theta + i\sin\theta)$. Describe geometrically the effect of multiplying z by a complex number of the form $z_1 = \cos\alpha + i\sin\alpha$, when $\alpha > 0$ and when $\alpha < 0$.

40. Suppose $z = \cos\theta + i\sin\theta$. If n is an integer, evaluate $z^n + \bar{z}^n$ and $z^n - \bar{z}^n$.

41. Write an equation that relates $\arg(z)$ to $\arg(1/z)$, $z \neq 0$.

42. Are there any special cases in which $\text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$? Prove your assertions.

43. How are the complex numbers z_1 and z_2 related if $\arg(z_1) = \arg(z_2)$?

44. Describe the set of points z in the complex plane that satisfy $\arg(z) = \pi/4$.

45. Student A states that, even though she can't find it in the text, she thinks that $\arg(\bar{z}) = -\arg(z)$. For example, she says, if $z = 1 + i$, then $\bar{z} = 1 - i$ and $\arg(z) = \pi/4$ and $\arg(\bar{z}) = -\pi/4$. Student B disagrees because he feels that he has a counterexample: If $z = i$, then $\bar{z} = -i$; we can take $\arg(i) = \pi/2$ and $\arg(-i) = 3\pi/2$ and so $\arg(i) \neq -\arg(-i)$. Take sides and defend your position.

46. Suppose z_1, z_2, and $z_1 z_2$ are complex numbers in the first quadrant and that the points $z = 0$, $z = 1$, z_1, z_2, and $z_1 z_2$ are labeled O, A, B, C, and D, respectively. Study the formula in (6) and then discuss how the triangles OAB and OCD are related.

47. Suppose $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$. If $z_1 = z_2$, then how are r_1 and r_2 related? How are θ_1 and θ_2 related?

48. Suppose z_1 is in the first quadrant. For each z_2, discuss the quadrant in which $z_1 z_2$ could be located.

(a) $z_2 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ **(b)** $z_2 = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$

(c) $z_2 = -i$ **(d)** $z_2 = -1$